

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

19 August 2016

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	20
2	11
3	12
4	12
5	13
6	12
Section B	
7 / 8	20
Total	100

This document consists of **18** printed pages.



Section AAnswer **all** questions.**1** A protease is an enzyme that digests protein.**(a)** With the aid of a diagram, describe the reaction catalysed by protease.

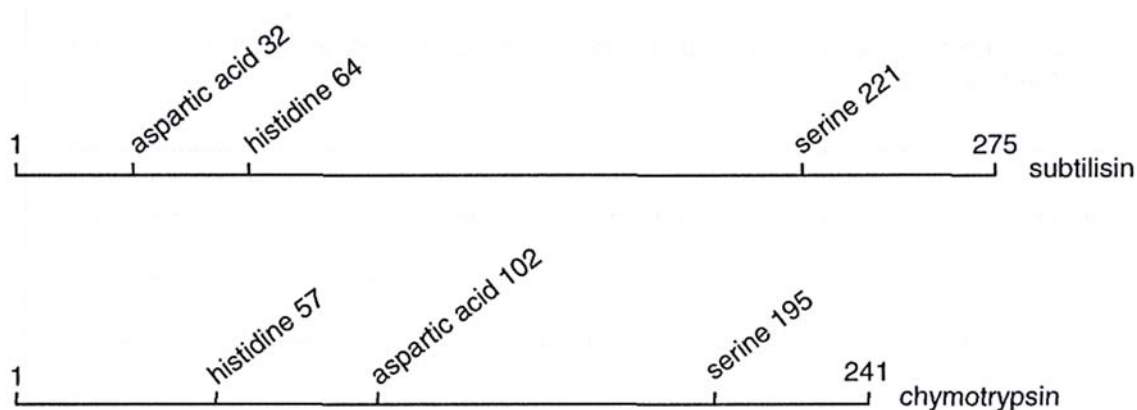
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..... [3]

Subtilisin is a protease synthesised by bacteria and is made up of 275 amino acids. Chymotrypsin is a protease synthesised by bacteria and is made up of 241 amino acids. Both enzymes have the same arrangement of three amino acids, serine, histidine and aspartic acid in their active sites but they are structurally different with the three amino acids being in different positions in the amino acid sequences shown in Fig. 1.1.

**Fig. 1.1**

- (b) With reference to Fig. 1.1 and your knowledge on levels of protein structure, describe how amino acid residues at different positions in the protein may be brought together in the active site.

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The graph in Fig. 1.2 shows how the activity of a protease varies with temperature.

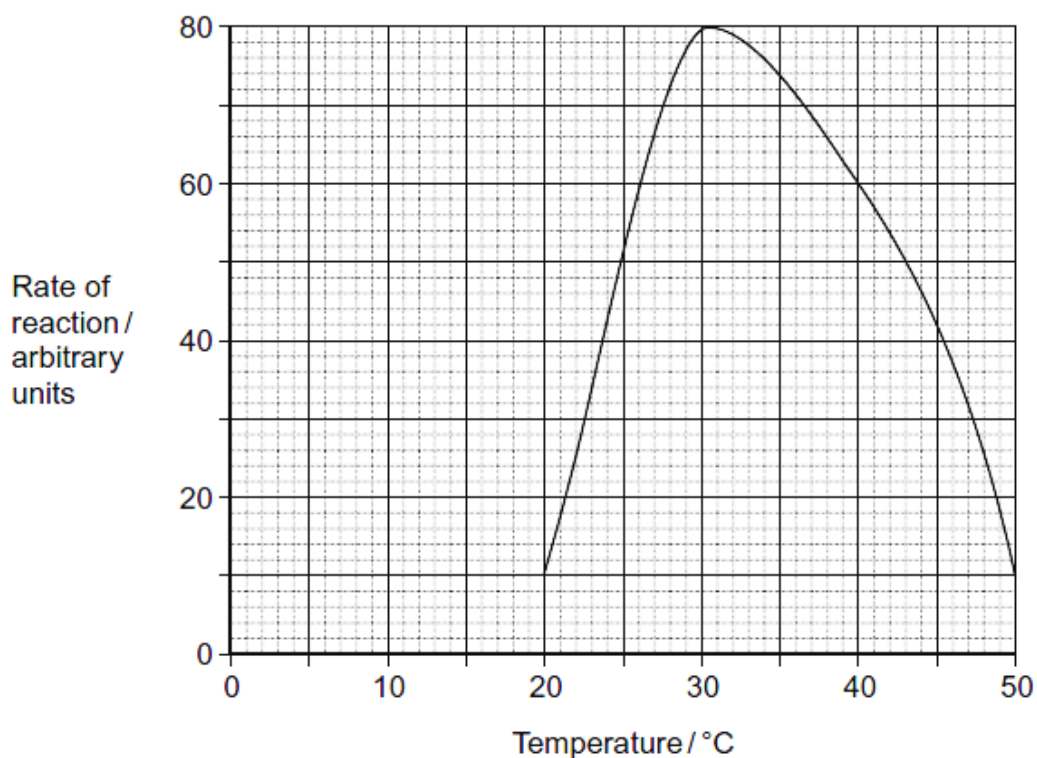


Fig. 1.2

- (c) With reference to Fig. 1.2, explain the shape of the graph between 30°C and 50°C.

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Students investigated the effect of pH on the activity of protease.

- The students used agar plates containing protein. The protein made the agar cloudy.
- They made four wells of equal size in the agar of each plate.
- They added a drop of protease solution to each of the wells. The protease solution in each well was at different pH.
- The students incubated the agar plates for 4 hours at constant temperature.

Fig. 1.3 shows the agar plates after they were incubated and the pH of the protease solution in each well.

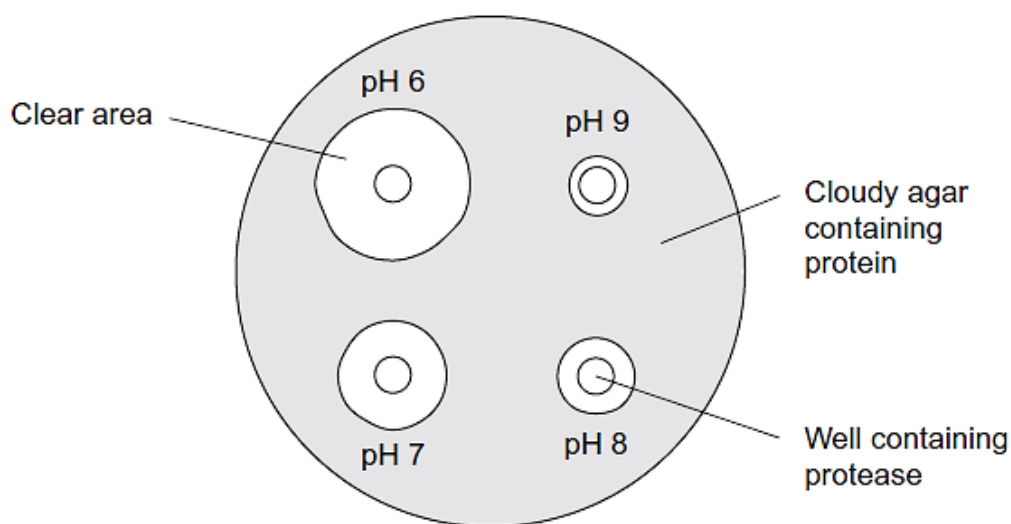


Fig. 1.3

(d) With reference to Fig. 1.3,

- (i)** suggest a suitable temperature for incubating the agar plates and explain your answer.

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- (ii)** explain the effect of pH on the activity of this protease.

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..... [1]

The gene encoding protease is transcribed to give rise to mRNA. The mRNA transcribed leaves the nucleus and is translated by ribosomes in the cytosol. Fig. 1.4 shows the process of translation.

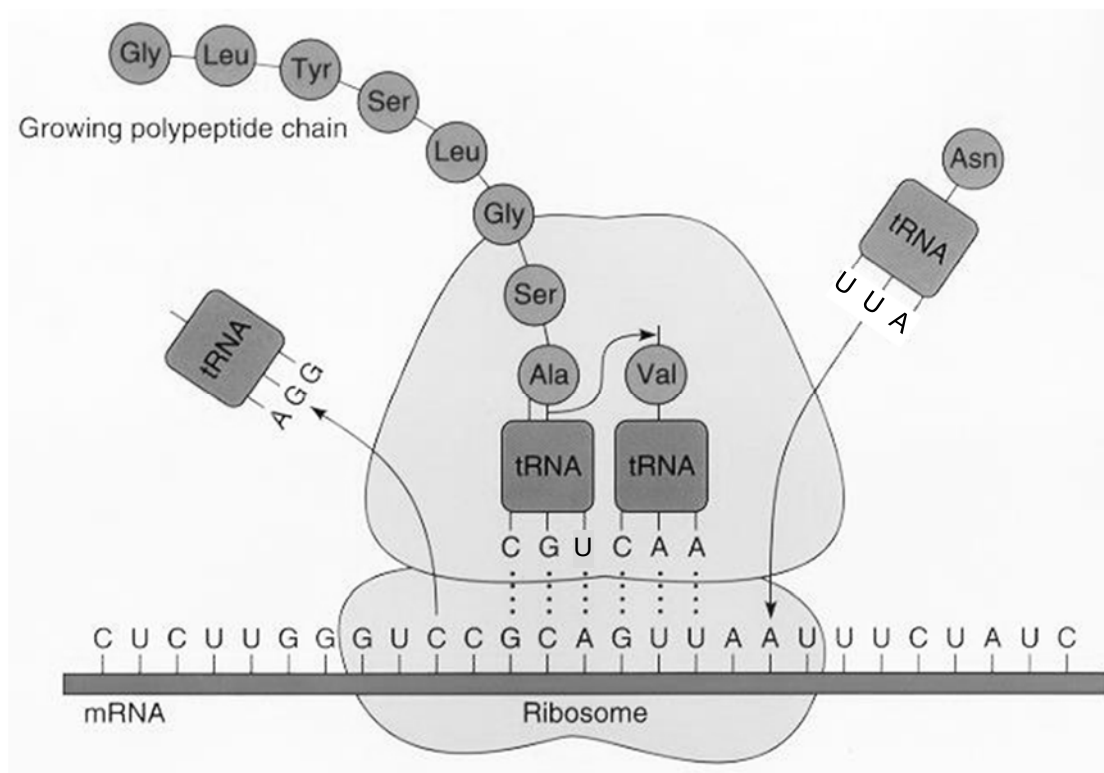


Fig. 1.4

- (e) With reference to Fig. 1.4, describe how the amino acid asparagine (asn) is incorporated into the growing polypeptide chain.

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[4]

Aminoacyl tRNA synthetase is an enzyme that catalyses the attachment of amino acids to its corresponding tRNA.

A specific aminoacyl tRNA synthetase allows the amino acid asparagine to bind to its active site. A single base substitution occurred in the gene coding for this enzyme, resulting in the enzyme recognising proline instead of asparagine.

Fig. 1.5 shows the effect of this mutation on the resultant aminoacyl tRNA.

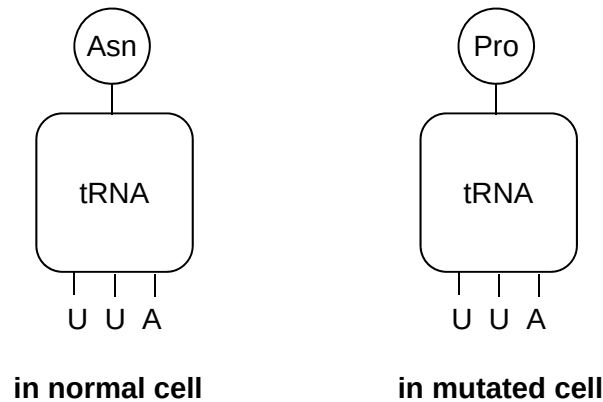


Fig. 1.5

- (f)** Explain how the structure of tRNA is adapted to perform its function.

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..... [2]

- (g)** Suggest why a mutation in this gene would have extremely damaging effects on the organism.

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[Total: 20]

- 2 Human Immunodeficiency Virus (HIV) is a retrovirus that causes acquired immune deficiency syndrome (AIDS).

(a) Outline the process by which HIV enters the host cell.

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..... [2]

Many integrase inhibitors have been discovered in recent years, and some of them are presently in clinical trials. One such inhibitor is Raltegravir.

In a clinical trial, one group of patients was given Raltegravir while the control group was given a placebo (drug-free pills) over 24 weeks. The number of copies of HIV-1 RNA per ml of blood plasma was measured and the results are given in Fig. 2.1.

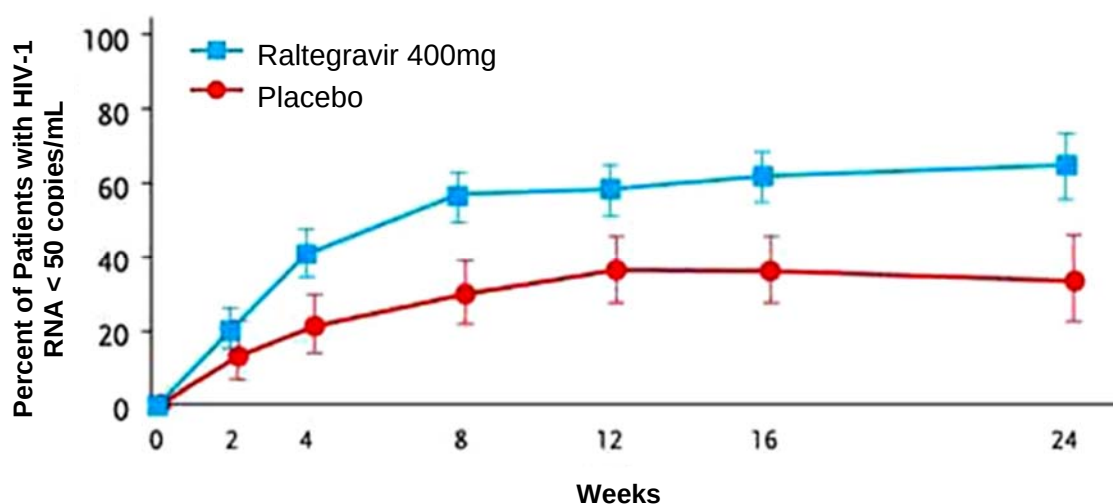


Fig 2.1

(b) With reference to Fig. 2.1, describe the difference in the effect of treating patients with Raltegravir and placebo.

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- (c) Suggest how Raltegravir works as an antiretroviral drug.

----- [2]

- (d) The viral genome undergoes frequent mutation.

Suggest why this means that the action of Raltegravir may no longer be effective in future.

----- [2]

Another important process in the reproductive cycle of HIV is the cleaving of polyproteins into functional proteins using protease. Fig. 2.2 shows such a process.

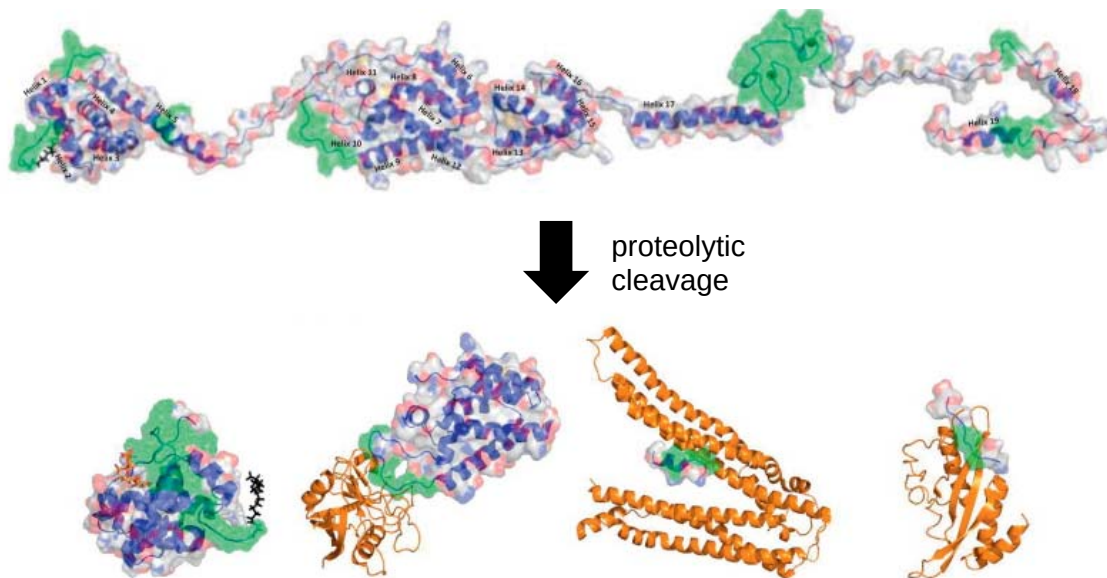


Fig 2.2

- (e) Describe the polypeptides and the products formed after proteolytic cleavage.

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[Total: 11]

- 3 Fig. 3.1 shows the structure of a nucleosome.

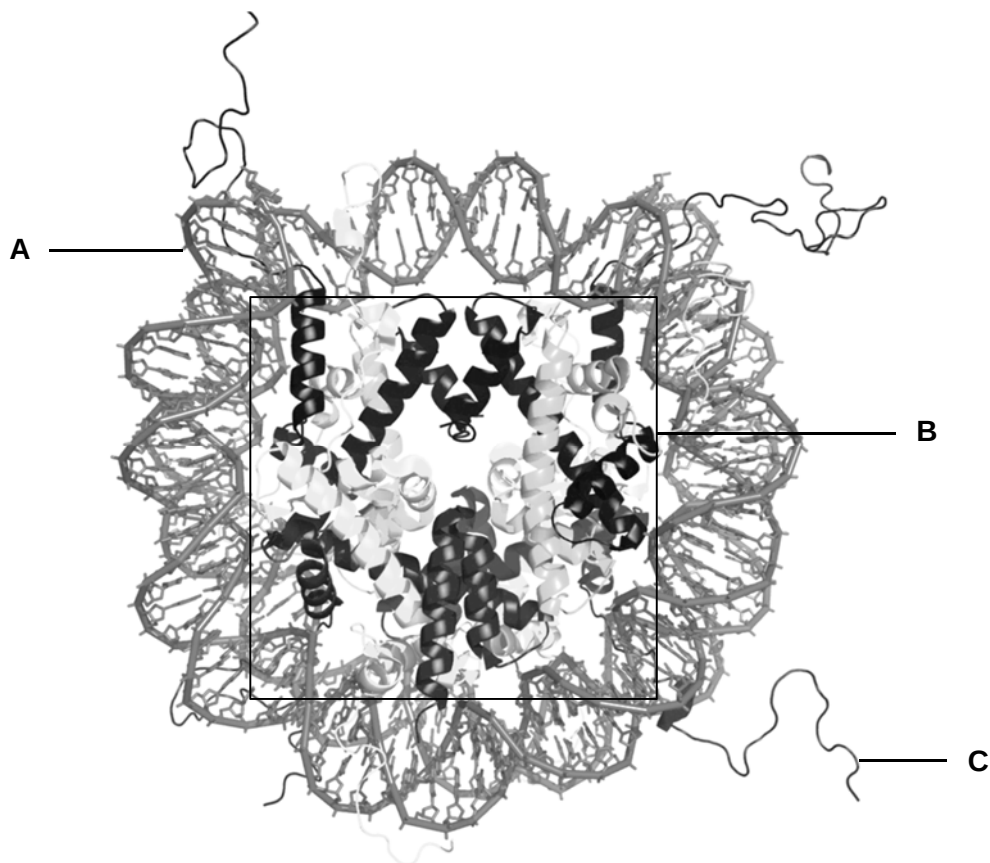


Fig. 3.1

- (a) With reference to Fig. 3.1,
- (i) label the structures A to C.

A

B

C

[2]

(ii) describe how structure **A** is stabilised.

----- [2]

(iii) state the secondary structure observed in structure **B**.

----- [1]

(iv) describe how structure **C** can be modified to influence gene expression.

----- [2]

(b) Describe how DNA is packaged in eukaryotes.

----- [5]

[Total: 12]

- 4 (a) Explain what is meant by *epistasis*.

----- [2]

In some species of plants, malvidin is a primary flower pigment that results in the colour wine red. Gene **A/a** codes for an enzyme that synthesises malvidin while gene **H/h** was discovered to inhibit its synthesis resulting in white flowers.

In one experiment, double heterozygote parents were crossed to produce F1 generation progeny comprising of 452 white flowering plants and 114 red wine flowering plants.

- (b) Using the symbols for the alleles stated above, draw a genetic diagram to show the expected phenotypic ratio for the F1 offspring.

[5]

- (c) Determine, using an appropriate statistical test, if the observed results are expected. The critical values for the chi-squared test are provided in Table 4.1.

Table 4.1

degree of freedom	Probability, p				
	0.1	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

[5]

[Total: 12]

- 5 (a) A myelinated axon transmits impulses faster than a non-myelinated axon.

Explain this difference.

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..... [2]

Doctors investigated the relationship between myelin in the brain tissue and different types of dementia. All types of dementia involve loss of mental ability.

The doctors measured the mean amount of myelin in samples of brain tissue from:

- a control group of 12 people without dementia
- 20 people with vascular dementia (VaD)
- 19 people with Alzheimer's dementia (AD)
- 31 people with Lewy body dementia (LD).

The doctors' results are shown in Fig. 5.1. The vertical bars show standard errors.

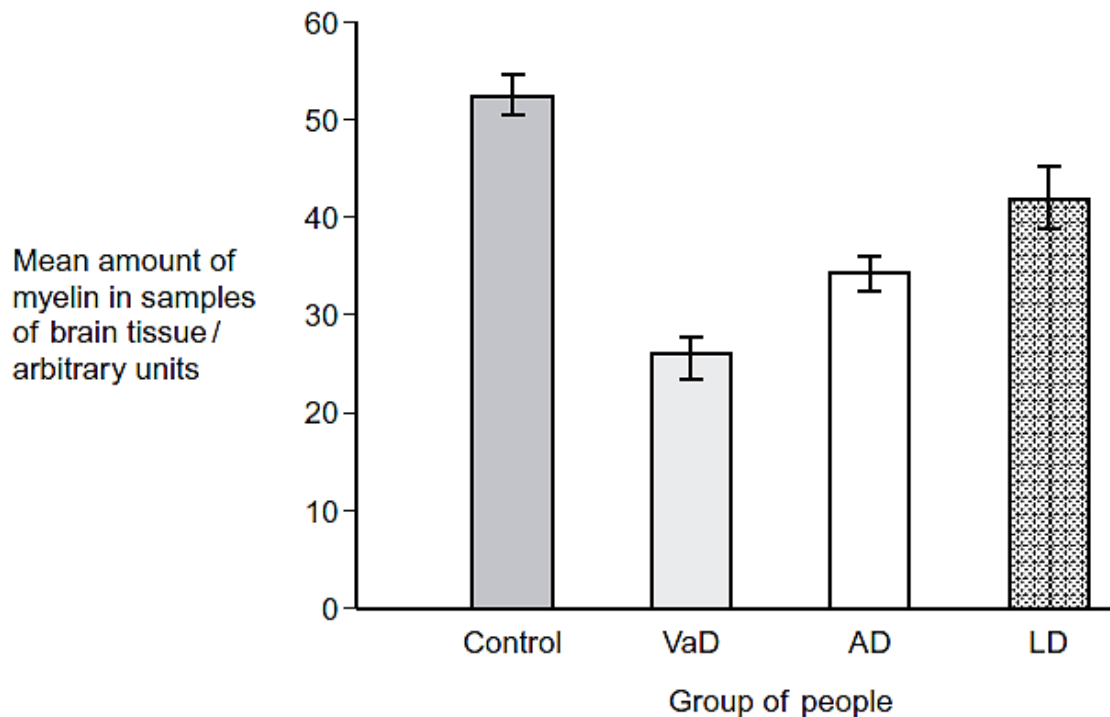


Fig. 5.1

- (b) With reference to Fig. 5.1,

- (i) explain the significance of the standard error bars.

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..... [1]

- (ii) discuss whether the data supports the conclusion that there is a relationship between incidence of dementia and the amount of myelin in a person's brain.

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Cocaine is a drug that interferes with synaptic transfer. It affects neurones that release dopamine, an excitatory neurotransmitter associated with feelings of pleasure.

Fig. 5.2 shows the mechanism of action of cocaine.

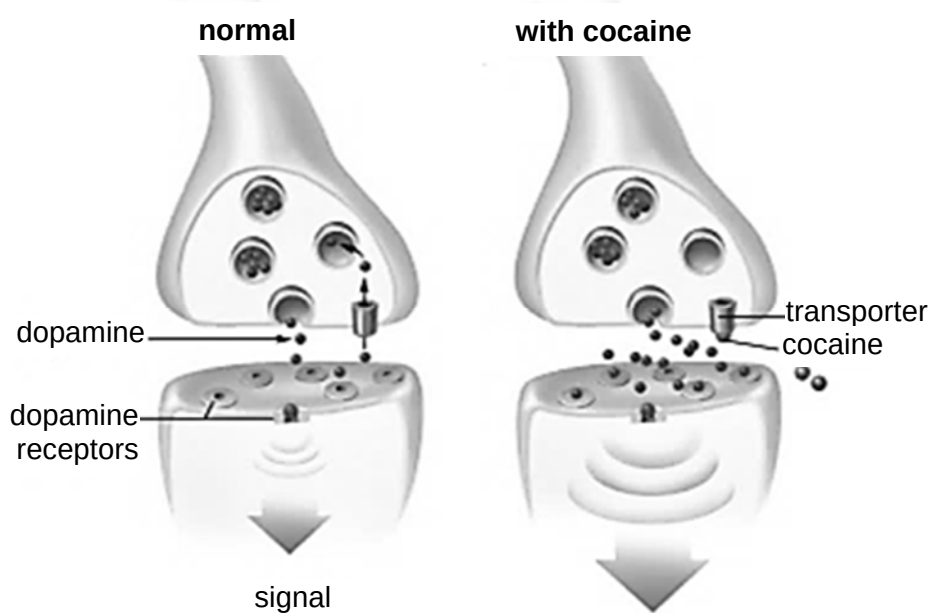


Fig. 5.2

- (c) (i) Describe the process of synaptic transfer in normal individuals.

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..... [4]

- (ii) With reference to Fig. 5.2, explain the effects of cocaine.

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- (iii) Long-term cocaine use results in a decrease in the number of dopamine receptors in the post-synaptic membrane.

Suggest why cocaine abusers are no longer able to feel pleasure naturally.

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..... [1]

[Total: 13]

- (a)** Explain how the ten species of fish evolved.

[illegible]

Suggest what might happen to the ten species of fish.

[2]

Tree frogs are commonly found in Southern Australia. *Hyla ewingi* and *Hyla verrauxi* are two closely related species of tree frogs. Research from breeding studies and DNA sequence data has shown that they have strong genetic compatibility.

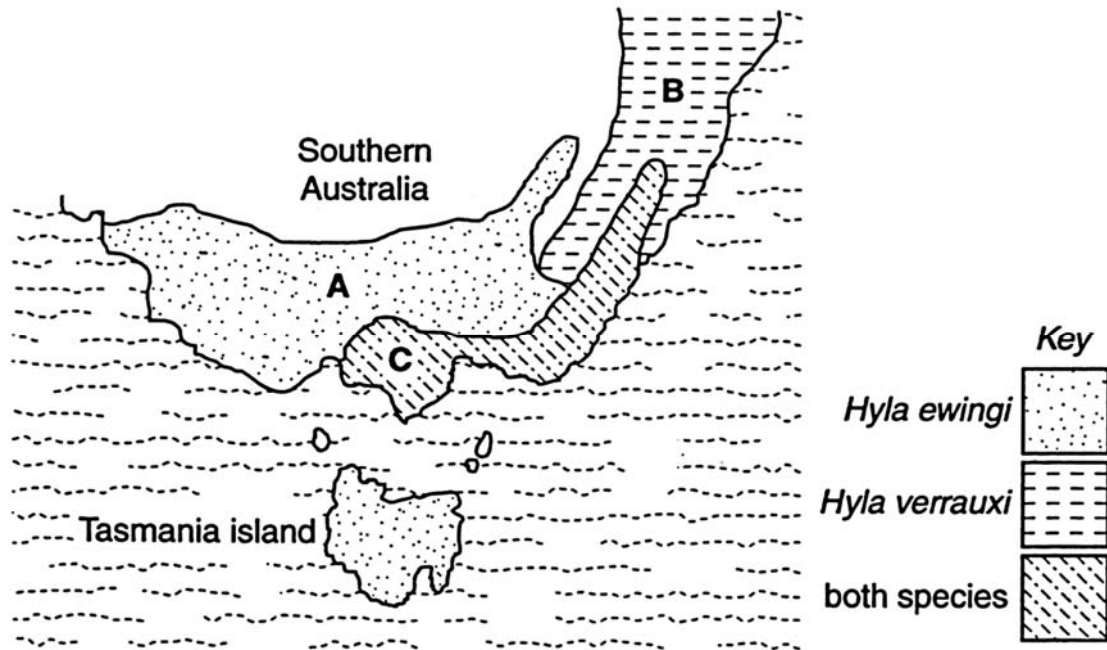


Fig 6.1

- (c) (i) State the genus of the tree frog.

[1]

- (ii) With reference to Fig. 6.1 and the research studies, explain how the scientists concluded that the two species of tree frogs are closely related.

[4]

[Total: 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

7 **(a)** Describe the main stages in Calvin cycle. [8]

(b) Describe the roles of membranes in photosynthesis and in respiration. [8]

(c) Explain how membrane fluidity is maintained in a cell when temperature increases. [4]

[Total: 20]

8 **(a)** Describe the structural features of collagen and how they contribute to its function. [8]

(b) Insulin polypeptide synthesised by ribosomes attached to the endoplasmic reticulum will be transported out of the beta cell.

 Describe the route taken by insulin out of the beta cell. [8]

(c) Explain the roles of vesicles that emerge from the Golgi body. [4]

[Total: 20]